

ABERDARE TECHNICAL COLLEGE

**UNIT: PERFORM COMPUTER OPERATIONS
ELEMENT: MANIPULATE COMPUTERIZED SPREADSHEET**

**UNIT CODE: 0611 351 02A
ASSESSMENT: PRACTICAL 1 TERM 2
TIME: 2 HOURS
TOTAL MARKS: 50 MARKS**

Instructions to the Candidate

1. Use Microsoft Excel or another approved spreadsheet application.
2. Create a new spreadsheet workbook.
3. Save the workbook as **Student_Spreadsheet_Practical.xlsx**.
4. Perform all activities as instructed.
5. Enter data accurately and use formulas/functions where required.
6. Save your work regularly.
7. Print the required worksheet(s) after completing the assessment.
8. Submit the completed workbook and printed output to the assessor.

You have been provided with the following materials

- ✓ A desktop computer installed with a database software (Access).
- ✓ Ensure everything is functional before the commencement of the test

TASK 1: CREATE AND ENTER DATA IN A WORKBOOK

Activity 1.1: Create the workbook

Open Microsoft Excel and create a **new workbook**.

Rename the first worksheet:

Sales Report

Create a second worksheet and rename it:

Summary

Activity 1.2: Enter the following data

In the **Sales Report** worksheet, enter the following information:

Product	January	February	March	Price per Unit
Computers	15	18	20	55000
Printers	12	15	17	18000
Monitors	20	22	25	15000
Keyboards	30	35	40	2500
Mouse	35	40	45	1500
UPS	18	20	24	6000

Add the following column headings:

- **Total Units**
- **Total Sales**

Activity 1.3: Save the workbook

Save the workbook as:

Student_Spreadsheet_Practical.xlsx

Activity 1.4: Demonstrate spreadsheet elements

Identify and demonstrate the following in your workbook:

- Worksheet
- Workbook
- Row
- Column
- Cell

TASK 2: CELL REFERENCING, FORMULAS AND FUNCTIONS

Using the **Sales Report** worksheet:

Activity 2.1: Relative cell referencing

In the **Total Units** column, calculate the total units sold for each product using a formula.

For example, for Computers:

January + February + March

Copy/fill the formula down for all products.

Activity 2.2: Absolute cell referencing

In an unused area of the worksheet, enter:

Item	Value
Tax Rate	16%

Calculate the **tax amount** for each product's total sales using the tax rate.

Use an **absolute cell reference** for the tax rate.

Activity 2.3: Mixed cell referencing

Create a small calculation table containing quantities and prices and demonstrate the use of a **mixed reference**.

Activity 2.4: Arithmetic functions

Use spreadsheet functions to calculate:

1. Total sales
2. Average sales
3. Maximum sales
4. Minimum sales

Use appropriate functions such as:

- SUM
- AVERAGE
- MAX
- MIN

Activity 2.5: Logical function

Create a column called **Performance**.

Use the IF function to classify products as:

- **Good** if Total Units are 50 or more
- **Needs Improvement** if Total Units are below 50

TASK 3: WORKSHEET EDITING AND DATA MANIPULATION

Activity 3.1: Insert rows and columns

1. Insert one new row above the existing data.
2. Insert one new column before the **Price per Unit** column.

3. Enter an appropriate heading in the new column.

Activity 3.2: Adjust rows and columns

Adjust the:

- Row height
- Column width

so that all headings and data are clearly visible.

Activity 3.3: Copy and move worksheets

Make a copy of the **Sales Report** worksheet.

Rename the copied worksheet:

Sales Report Copy

Move the copied worksheet so that it appears after the **Summary** worksheet.

Activity 3.4: Delete worksheet

Delete the **Sales Report Copy** worksheet.

Activity 3.5: Data editing

Perform the following:

1. Change the Computers January sales from **15 to 17**.
2. Change the Mouse March sales from **45 to 48**.
3. Add a new product called **Flash Disk** with appropriate sample sales figures.

Activity 3.6: Insert a worksheet

Insert another worksheet and name it:

Lookup

TASK 4: LOOKUP FUNCTIONS AND WORKSHEET FORMATTING

Activity 4.1: Create a lookup table

On the **Lookup** worksheet, enter:

Product	Category
Computers	Hardware
Printers	Hardware
Monitors	Hardware
Keyboards	Accessories
Mouse	Accessories
UPS	Power Backup
Flash Disk	Storage

Add a **Category** column to the Sales Report worksheet.

Use an appropriate lookup function such as **XLOOKUP** or **VLOOKUP** to automatically retrieve the category for each product.

Activity 4.2: Apply formatting

Format the **Sales Report** worksheet as follows:

1. Make the main headings **bold**.
2. Apply an appropriate font style and size.
3. Centre-align column headings.
4. Apply borders to the data table.
5. Apply shading to the heading row.

6. Adjust column widths appropriately.

Activity 4.3: Header and footer

Insert a header containing:

PERFORM COMPUTER OPERATIONS – SPREADSHEET PRACTICAL

Insert a footer containing:

Candidate Name and Page Number

TASK 5: CHART GENERATION AND WORKBOOK PRINTING

Activity 5.1: Create a chart

Using the **Product** and **Total Units** data, create a suitable **column chart**.

Give the chart the title:

Total Units Sold by Product

Activity 5.2: Label and edit the chart

Edit the chart by adding:

- Chart title
- Horizontal axis title: **Products**
- Vertical axis title: **Units Sold**
- Data labels

Position the chart appropriately on the worksheet.

Activity 5.3: Print setup

Configure the worksheet for printing:

1. Set the appropriate page orientation.
2. Set suitable margins.
3. Ensure the table and chart fit appropriately on the printed page.
4. Set an appropriate print area.

Activity 5.4: Print

Print the completed **Sales Report** worksheet.

The end of practical assessment, the trainer is introducing oral questions shortly